

Semantic Segmentation of Cancerous Tumours in Histopathological Images

Abstract

Deep learning semantic segmentation models have garnered considerable attention and growth in medical image analysis, specifically tumor identification and annotation in histopathology. There have been rapid advancements in this field over recent years with new models being proposed that each provide new state-of-the-art performance. However, there is still a struggle to balance local and global feature extraction, with models often biased towards one of the two. The medical domain also threatens new deep-learning models that have previously utilized vast amounts of data to improve accuracy and generalisation. Medical data is limited in size, often poorly annotated, includes human bias, and is limited in variety, thus stalling modern models' full potential and generalizability. Vision transformers and self-supervised learning have emerged as tools that could effectively resolve these issues. This paper aims to understand previous and current state-of-the-art approaches, vision transformers, and self-supervised learning to propose a novel multiscale vision transformer framework that effectively resolves these issues.

✓ I have not used any GenAI tools in preparing this assessment.

✓ I certify that all material in this dissertation which is not my own has been identified.

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1 Introduction

Medical image tumor analysis is the detailed identification, annotation, and segmentation of potential cancerous tumors in medical images and is used by medical professionals to support cancer diagnosis [1]. Histopathological images are high-resolution intricate images that help highlight cell structures and are used to analyse potentially cancerous tumors. They are produced in a difficult time-consuming process, where $3-5\mu m$ thick cell frames are prepared, stained and digitized [1]. Yet, histopathological images are not easily analysed by pathologists since they contain complex unclear boundary/semantic regions [2] and heterogeneous qualities after staining [1]. Deep learning (DL) models have become predominant tools in performing/aiding medical image analysis, particularly semantic segmentation models. Semantic segmentation mimics the role of medical image analysis, clustering together and isolating pixels in an image that belong to the same object class [3]. It can be more formally defined as pixel-wise classification [3]. DL methods have proved themselves superior to human annotation, providing reduced processing times and quantitative measures in images [4]. However, many significant challenges in this field still need to be overcome until perfect systems can be realized.

The current medical image analysis domain faces significant challenges. Privacy laws, database restrictions, and the costly limited output of new histopathological images severely restrict the amount of data that can be used to train models; consequently, impacting the generalizability and accuracy of models [3, 5]. Style and technique inconsistencies across histopathological images hinder effective feature map learning since datasets are more likely to contain noise, class imbalances, and incorrect annotations [3, 4]. Modern state-of-the-art DL segmentation models such as FCN [6] and U-Net [7], though influential, struggle to incorporate global contextual information in segmentation outputs; hindering the effectiveness of tumor segmentation in histopathological images where broader spatial relationships are crucial for accurate segmentation [8, 9]. The binary segmentation problem threatens model learning, causing models to learn background and foreground relationships rather than image features, resulting in ineffective models that have become overfitted to training data [3].

In this literature review, I will analyse previous semantic segmentation frameworks, identifying effective qualities that can be integrated into a novel multiscale vision transformer-based framework to determine whether or not multiscale vision transformers can enhance the ability to capture spatial/contextual information, improve the segmentation of tumors with different morphologies, and improve model generalizability in a domain known for poor data availability and quality.

2 Preliminary Research

2.1 Early Approaches

Pathologists manually analysed and annotated complex histopathological images by hand [10]. Operating in supervised pairs, pathologists applied numerous post-processing steps to achieve successful analysis [1, 11], such as fat-dissolving, which removed unnecessary fat around clustered cells or small tumors (less than $5mm$ in diameter) [11, 10]. Manual annotation was time-consuming, visually fatiguing, prone to human error, and had strong human bias [2, 12]. Moreover, the ineffectiveness of manual annotation, couldn't meet current workloads/demands in modern healthcare services, leading to delayed, incorrect diagnosis, and worsened patient outcomes [11].

Pathologists progressed to manually creating matrix feature maps applied in a sliding window process that highlighted tumors [3]. Labour-intensive and requiring expert pathologists, this approach struggled to capture fine-grained information or hierarchical structures [3] and had poor generalizability since it was infeasible to train on large datasets [4]. Thus, it provided little support to pathologists.

Semi-automatic approaches were later introduced, where regions of interest or boundaries in histopathological images, were proposed by pathologists and segmented by convolutional neural networks (CNN) [13]. However, semi-automatic models struggled to process large crops of datasets since pathologist region proposal output was low and susceptible to human bias [13].

DL models marked a significant jump in medical image tumor analysis [12]. Automatically and independently identifying regions of interest they reduced segmentation inference time [3] and human-bias. End-to-end learning removed training model components separately, supporting efficient learning of complex latent representations in a single pipeline process [6, 7]. Feature maps could be automatically inferred from training data using highly robust hierarchical feature map learning [3, 6, 7]. DL semi-supervised and self-supervised learning strategies improved medical data availability. Where, semi-supervised learning creates pseudo training labels from unlabelled data [5], and self-supervised learning extends the usability of training data by performing pretext tasks to learn intrinsic qualities and features of input images [5].

2.2 U-Net

Building upon Fully Convolutional Neural Networks (FCN) [6], U-Net [7] positioned itself as a specialised encoder-decoder medical image segmentation model focused on achieving robust high-resolution detailed segmentation outputs without requiring large well-annotated datasets, something FCN and other medical image segmentation models had previously struggled with [7, 6].

U-Net introduced skip connections, which propagated essential high-resolution information across the network, concatenating it with deeper semantic information, encouraging the network to re-use shallow layer detailed representations and deeper layer image-level context for accurate predictions that captured both spatial context and wider global context [14, 15].

U-Net pioneered training data augmentations such as rotations and colour variations, to increase the size and variety of inadequate medical datasets [7, 5]. Therefore, improving invariance learning and model generalizability, since features are learnt invariant to changes, which enables better performance on unseen data and heterogeneous tumors [7]. However, the scope of impact was limited, as "*real-world variety*" was hard to create in inadequate datasets, which were most likely derived from a small pool of patients with the same imaging equipment/techniques.

Making significant progress in the medical image segmentation domain U-Net was now considered the backbone framework for medical tasks, replacing and automating many redundant and specialized tasks [14]. However, U-Net still suffered to accurately represent detailed global contextual features, even with the integration of skip connections.

2.3 Vision Transformers (ViT) for Semantic Segmentation

Transformers are regarded as the state-of-the-art methodology for NLP tasks [16], but have recently gained attention in computer vision tasks [17]. Discarding traditional convolutional layers that struggled to capture global structures ViTs are built using self-attention mechanisms [16, 17]. Self-attention mechanisms effectively capture long-range dependencies, and global context information, and help avoid underlying feature loss [18].

2.3.1 Patch Embeddings

Images are divided into a sequence of 2D non-overlapping patches [17], which are flattened and linearly embedded into a constant latent vector size D , creating a sequence of *patch embedding* vectors that represent parts of an image. Since computer vision tasks require spatial information to determine the "*big picture*", *patch embeddings* are combined with learnable *positional encodings*, helping represent spatial relationships between image patches. Patch size is crucial in determining the level of detail and scope of global context the model learns; smaller patches capture fine details, while larger patches capture global context [19]. However, smaller patch sizes increase the quantity of *patch embeddings*, significantly impacting ViT computation and memory demands quadratically i.e. $O(patch^2)$, as there are more patches to attend together [17]. Moreover, histopathological images increase computational and memory demands, as they are high-resolution, produce long patch sequences, and require small patches to capture fine details and intricate global relationships. However, computational needs can be balanced by utilizing parallelized processes and limiting attention computation [17, 20, 21].

2.3.2 Attention, Self-Attention, and Multi-Head Self-Attention

Attention mechanisms enable sequence and dependency modelling without requiring input distance/sequence, allowing relevant input components to be focused on even if they are far apart [16]. Therefore, supporting global information integration across the entire images [17]. Attention mechanisms can be formally described as "mapping a query and a set of key-value pairs to an output, where the query, key, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key" [16]. Essentially describing how attention computes the compatibility between different patches using queries and keys, that represent higher-level information, to derive the presence and importance of features - patches attend to each other.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where $Q = \text{patch} \cdot W^Q, K = \text{patch} \cdot W^K, V = \text{patch} \cdot W^V$

Figure 1: Scaled Dot Product Attention [16]

Self-attention extends traditional attention mechanisms, deriving importance directly from a single input space. Patch embeddings attend to each other, determining the presence of object boundaries and distinctive features via compatibility. *Patch embeddings* are linearly projected into matrices Q (Query), K (Key) and V (Value). Q represents the focus of *patch embeddings*, K represents compatibility information about *patch embeddings*, and V represents the actual contents of *patch embeddings*. Q , K and V are computed using learnt weights. Raw attention scores are computed using matrix multiplication between Q and K , determining present compatibilities between each *patch embedding* (Figure 1). Attention scores are then scaled down, normalized, converted to a probability distribution summing to 1 using *softmax*, and multiplied with V to derive new patch representations (Figure 1).

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

Parameter matrices: $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}, W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k},$
 $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}, W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$

Figure 2: Multi-Head Attention Mechanism [16]

Multi-head self-attention further extends traditional attention mechanisms, focusing on capturing diverse latent representations of *patch embeddings* via multiple parallel self-attention heads. A main attention head projects patches into matrices Q , K , and V , which are fed into h attention heads, each with their own set of learnable projection matrices for Q , K , and V (Figure 2). Attention heads independently derive new representations of V , exhibiting a specific set of diverse latent representations. These representations are then concatenated together, creating a unified representation of features across all heads, before a final learnable projection is applied, transforming V_{combined} back into the original embedding dimension space (Figure 2).

Multi-head attention enables efficient parallelized computation of multiple latent feature representations [16, 17], which is beneficial in histopathology where, as previously discussed in **section 2.3.1** computational requirements are high. However, it is also the case in histopathology that attention mechanisms may perform redundant computation. Tumors and cell groups are often in distinct medium-sized groupings and thus cells are more likely to relate to neighbouring cells than cells far away [1]. Therefore, patches close together in positional encodings are more likely to provide valuable attention computation than patches far away. Consequently, computing attention across the entire patch space is redundant. Boundary-defined attention mechanisms like that in Swin Transformer [20] could reduce computation further by limiting attention to a smaller subset of patches that are potentially more relevant to each other.

2.3.3 Encoder

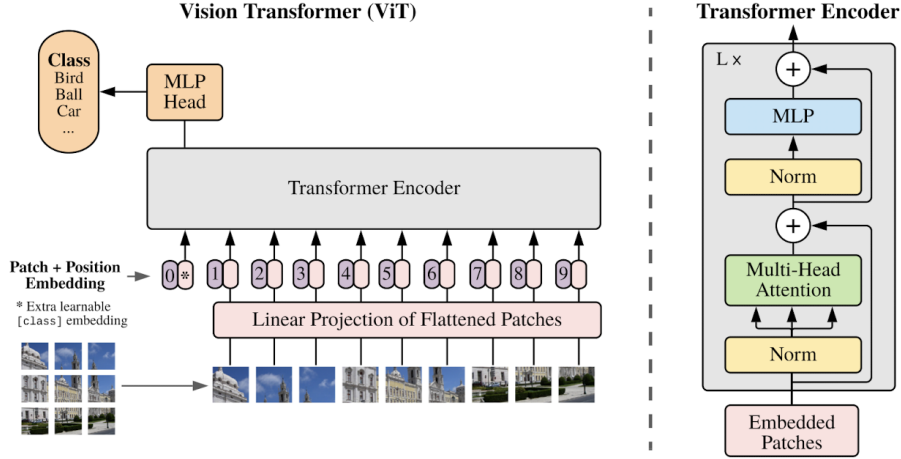


Figure 3: Classification Vision Transformer and Encoder Block [17]

ViT encoder blocks combine multi-head self-attention and *patch embedding* mechanisms into a coherent feed-forward pipeline. Encoder blocks contain a repeated collection of operations that progressively refine and transform image features through attention and non-linear transformations (Figure 3). *Patch embeddings* are first normalized, avoiding overactive features, and stabilizing future model training. A multi-head self-attention layer then extracts relationships, features, and local/global representations from *patch embeddings*. ViTs introduce the aforementioned skip connections (section 2.2) in the form of residual connections, which smooth gradient flow by propagating information from previous layers with newly derived features. A residual connection combines the original unnormalized *patch embeddings* with their new representations after multi-head self-attention, facilitating the propagation of previous features and long-range dependencies, while also counterbalancing the impact of significant changes after attention. *Patch embeddings* are normalized again to further stabilise attention, before being fed into a multilayer perceptron layer (MLP). The MLP layer further enriches features by expanding it with multiple linear projections and performing non-linear transformations using GELU activation to refine features [22]. A final residual connection is used combining the previous residual connection output with the new feature representations produced by the MLP layer, further stabilising attention and propagating long-range dependencies. Multiple transformer encoder blocks can be stacked together in successive layers, creating a deeper model that evolves *patch embeddings* gradually to extract increasingly abstract and latent features/representations.

2.3.4 Decoder

Segmentation Transformer (SETR) [23] proposed a simple progressive upsampling decoder architecture that reshaped, combined, and gradually upsampled *patch embeddings* into their original spatial dimensions. *Patch embeddings* are transformed and combined into a single 2D feature map, which is fed into multiple bilinear upsampling layers that aggregate and reduce feature dimensions while increasing spatial dimensions. SETR progressively upsamples feature representations like in FCN and U-Net, limiting segmentation noise since the resolution is increased incrementally rather than all at once. SETR also proposed a multi-level feature aggregation decoder which employed skip connections to upsample and combine opposite encoder-decoder block feature representations [23]. Therefore, restoring feature representations and spatial dimensions during feature upsampling, and enabling more accurate outputs. However, SETR used a single static sized *patch embedding*, which limited its ability to capture multiple scales of context from high-resolution images.

Fully transformer-based decoder architectures can also be utilised to upsample *patch embeddings*, one such model is Segmenter [24]. Segmenter signified that skip connections could easily integrate into transformer-based architectures as there was no need to translate and upsample outputs like that in SETR [23]. Segmenter and other models [25] have incorporated this form of decoder to naturally

propagate early representations throughout the network, improving feature representation integrity, and reducing vanishing gradients.

2.3.5 Recent Advances in Multiscale Vision Transformer

Since ViTs operate with large receptive fields capturing global representations they inherently lack the inductive bias present in CNNs [17] and struggle to effectively capture local fine details, which is critical in histopathological images; where lack of tumor detail and small tumor identification can hinder accurate diagnosis [11]. Multiscale ViT approaches have presented possible solutions that effectively capture local features while maintaining global context extraction, without exploding computation and memory requirements.

The Swin Transformer proposed a novel approach that increased local feature extraction, maintained global context consideration, and lowered computational demands [19]. Patch size was reduced to 4x4 from 16x16 to more easily capture fine detail and attended to by a shifted window self-attention mechanism. Each layer aggregates patches in a 2x2 radius into a single feature-rich patch representation, before shifting the window boundaries to encapsulate new patch combinations. Reducing patch size traditionally increases computation as each patch had more patches to attend to, but increases local feature and fine-detail extraction as receptive fields are smaller. Shifted window self-attention creates window boundaries around patches, reducing computation as attention is calculated in smaller parallelized groups. Patch aggregation combines patches and increases feature channels to effectively create broader patches that integrate many smaller contextual relationships. Therefore, building an increasingly more global view of patches that are then attended to identify wider global relationships. Furthermore, Swin Transformer’s pyramid-like structure enabled local fine-grained relationships to be captured in lower levels using smaller receptive fields, which are then aggregated and coarsened in deeper layers to contextualize global relationships. Moreover, shifted window self-attention could be highly beneficial in local and global feature extraction in histopathological images that often contain many groups of clustered cells [1], as attention is focused on potentially more relevant areas.

Multiscale can also be achieved using different sized *patch embeddings* in parallel [21]. Thus, allowing small, medium, and large patches, each representing varying levels of fine detail and global context, to be processed independently and combined into a comprehensive feature representation. HRViT [26] is an example model that incorporates this. Consisting of multiple encoder block stages, HRViT maintains a pool of different patch sizes and progressively in each new stage introduces new patch sizes that are derived from exiting patches [26]. Feature fusion is used during changing stages where smaller patches propagate up the patch hierarchy and are fused with the next largest patches and vice-versa [26]. Therefore, enriched smaller representations contribute to more context-representative patches and vice-versa, which improves local and global feature extraction as patches capture different scaled representations of features and progressively support each other in refining them.

2.4 Self-Supervised Learning

Traditionally model generalizability is improved using vast amounts of diverse data during training [5]. However, medical datasets cannot facilitate this [5]. Self-supervised learning (SSL) is a potential approach to improve ViT model generalizability in the presence of small datasets [19]. Using multiple diverse pretext tasks ViTs are encouraged to learn features with different goals in mind, resulting in invariant feature learning and reduced overfitting since datasets are used for numerous use cases [5]. SSL also helps avoid the binary segmentation problem, as models are encouraged to learn features and not irrelevant relationships present in the data. Therefore, improving model generalizability as learned representations are robust, transferable, and not prone to a task-specific basis.

Contrastive SSL operates on the principle of learning to differentiate similar (positive pairs) and dissimilar (negative pairs) representations. In the context of ViTs, this translates to the similarity and dissimilarity between *patch embeddings*. LETCP [25] proposed a novel contrastive SSL approach. Positive pairs were derived from each patch using different data transformations i.e. rotation, crops,

and colour change, and negative pairs were represented as the combination of different positive pair patches. Positive pairs would be fed into transformer encoder blocks, producing *patch embeddings* that emphasised specific features. An InfoNCE loss [27] was then employed to determine the contrast between processed *patch embeddings* and unprocessed negative pairings. In doing so, the vision transformer learnt feature similarities present in positive pairs, which effectively pre-trained the network before further fine-tuning.

Masked region SSL operates on the principle of randomly removing patches in an image and predicting the missing patches in the final output layer, encouraging the model to learn representations using limited structures, patterns and contexts [28]. In ViTs this translates to randomly removing *patch embeddings* before encoding. MAE [29] proposed a masked region process where 75% of patch embeddings are masked out and only used in the decoder ViT block, while the other 25% unmasked *patch embeddings* are fed into the ViT encoder block. These two sets of *patch embeddings* then recombine in the ViT decoder block, where unmasked patches would be enriched using their positional encoding and other preprocessed unmasked patches. Therefore this form of generative SSL improves a model’s understanding of data distribution and relationships as it encourages it to reconstruct information with only a partial view of the world. Furthermore, enhancing generalizability, robustness, and flexibility to noise and missing components.

2.5 Data Augmentation

ViTs are known for being data-hungry and require a large amount of data until an acceptable level of accuracy is reached [30]. Data augmentation increases the size of datasets by diversifying them and could help facilitate ViT data needs in the presence of small inadequate medical datasets [5]. Traditional methods include augmenting image quality (sharpness, blur, noise), appearance (brightness, saturation, contrast), and layout (rotation, scaling, deformation) [5]. Incorporating these methods exposes the model to more varieties of the same feature during training, which reduces overfitting as feature identification is more flexible [5]. Therefore, improving feature extraction, generalizability, and robustness as models become more familiar with feature variety. However, augmentations such as scaling and deformation may not benefit histopathology as features may be translated into unrealistic representations that confuse training.

3 Design Brief and Hypothesis

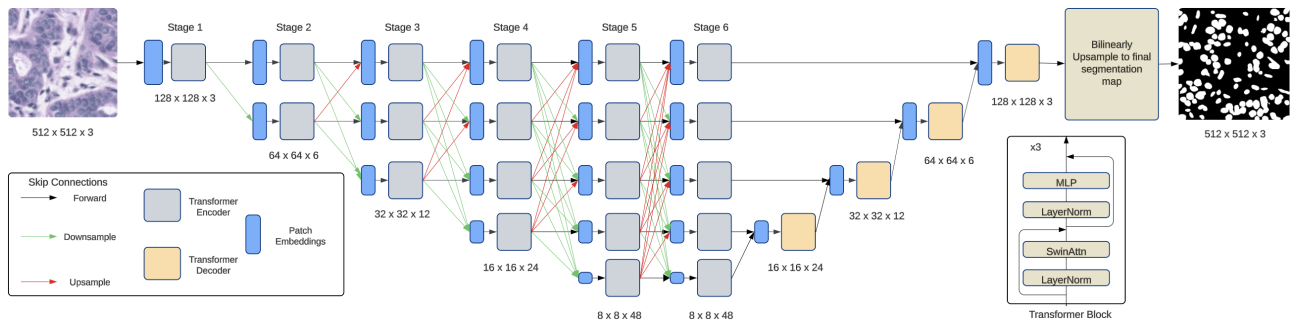


Figure 4: Novel HRViT and Swin Transformer Inspired Multiscale Vision Transformer Architecture

Previous CNN and ViT models struggled to capture a balance of local/global features and were often biased towards one of the two. To improve and unify global and local feature extraction, I propose a novel fully-transformer-based multiscale ViT architecture, shown in Figure 4. Building upon HRViT’s multiscale architecture I’ve replaced attention mechanisms with Swin Transformer attention and incorporated multiple progressive skip connections in the final decoder section. CNN components are limited since they hinder global context and require additional skip connection overheads. Leveraging HRViT as my foundation provides a superior multiscale feature hierarchy where multiple patch sizes (128x128 to 8x8) are processed simultaneously in a closely linked progressive and supportive skip connection network. HRViT’s multiscale patch network promotes wider context inference from

larger patches (128x128, 64x64, 32x32) and finer detail inference from smaller patches (16x16, 8x8) in a tightly coupled network where newly refined features propagate throughout the network providing additional contextual and fine-detailed information to all other patches - emulating zooming in and out on an image. HRViT’s various feature representation scales are combined, providing a complete picture that enhances spatial and contextual feature extraction of potentially heterogeneous features, i.e. tumors. Moreover, combining different scales in histopathology is critical in determining tumor features, where both fine detail and wider context are required for accurate tumor annotation [31]. High-resolution images pose significant computational and memory challenges for traditional ViTs, which limits their transferability to other domains. Swin Transformers have replaced HRViT attention mechanisms with a more bespoke and optimal attention mechanism that will provide a more focused and relevant attention calculation than other models that applied attention to all patches. As previously mentioned in **sections 2.3.2** and **2.3.5**, Swin Transformers’ attention boundaries could complement histopathology cell patterns, removing redundant attention computation, and potentially improving convergence to accurate outputs as patches are more likely to provide feature-relevant information during attention. Attention boundaries could also significantly reduce the computational demands posed by 512x512 high-resolution histopathological images, as each boundary region can be attended to in parallel, rather than one large process attending all patches together. I have chosen to incorporate skip connections into my architecture as they have proven to be an effective and easy-to-integrate method employed by multiple state-of-the-art encoder-decoder models (U-Net, SETR, Segmenter), to propagate, preserve, and improve spatial/contextual information throughout a model network. Together, these components will complement each other in enhancing spatial and contextual information extraction. The novel architecture provides multiple views of features from HRViT that are progressively refined and combined using tailored efficient Swin Transformer attention mechanisms, while decisive skip connections reinforce deeper layers with earlier feature representations and vice-versa, thus avoiding feature loss.

My proposed model like that of U-Net provides multiple mechanisms that can capture heterogeneous features present in data but cannot explicitly improve the generalizability of a model without some changes being made to the training process. As previously discussed in **section 2.4**, SSL is an effective method to train and improve the generalizability of models via pretext tasks. MAE masked region and LETCP contrastive SSL will be implemented to improve the generalizability of my model. LETCP contrastive SSL encourages the model to learn the difference between images, which could potentially contain different levels of tumors, therefore the model learns an understanding of what a tumor does and doesn’t look like. MAE masked region SSL will encourage the model to be robust in the face of noise and inconsistencies present in data, thus, improving performance and generalizability on unseen real-world datasets, which often contain these inconsistencies. Furthermore, both methods can help enhance model robustness against noisy and inconsistent data, which is often found in real-world data [5]. I will be using the SNOW dataset [32], a synthetically generated dataset containing 20,000 tumor-labelled histopathological images to present my model with an unbiased foundation for training. I also intend on utilising rotation and colour normalization data augmentation like in U-Net to increase the size, variety, and staining consistency of the data. Colour normalization will be maintained during future model inference to remove staining inconsistencies caused by pathologists when creating histopathological images [1].

3.1 Aims and Objectives (in order of priority)

1. Firstly, in line with my hypothesis the proposed ViT model should have enhanced accurate spatial and contextual feature extraction. Therefore, I expect model outputs to contain comprehensive fine-detail and global context, that is greater than that of another output from a model trained on the same data i.e. U-Net. I will compute multiple metrics (Intersection over Union (IoU), Dice Coefficient (Dice), Precision and Recall (P&R), and Hausdorff Distance (HD)) between each model output and the ground truth, to determine if my model provides a more comprehensive results than the other.

2. Model generalizability is a key component of my hypothesis and significantly determines how effective my model will be in the real world - a crucial quality. I have combined a feature hierarchy network with SSL and data augmentation to improve model generalisation. Since I will be training my model on synthetic data, the model will be unaware of real-world feature representations. Therefore, to evaluate this, I will assess my model against many real-world datasets (BreCaHAD [33], etc...) of varying quality using the previous metrics. IoU, Dice, P&R should be > 0.7 and HD should be < 10 .
3. Tumors are heterogeneous creating a constantly changing landscape, which makes analysis a constant struggle. As part of my hypothesis, I aimed to create a multiscale model that is designed to capture these qualities. To evaluate this, I intend to scale and deform histopathological images to create new formations of tumors. Manual inspection of attention maps during inference will be used to assess how the model distributes feature identification, with the hopes that smaller/larger features are driven by smaller/larger patches. Metrics IoU, Dice, P&R and HD will also be calculated to determine how capable the model is at identifying these heterogeneous features.

3.2 Timeline (Gantt Chart)

Milestone	Deliverable	Dec		Jan		Feb		Mar		Apr	
		1w-2w	3w-4w	1w-2w	3w-4w	1w-2w	3w-4w	1w-2w	3w-4w	1w-2w	3w-4w
Dataset preparation	Source all datasets										
	Pytorch datasets "DataSet" and "DataLoader" classes										
ViT Model	ViT Encoder component										
	ViT Decoder component										
	ViT Segmentation output component										
SSL	Contrastive integration with ViT										
	Masking region integration with ViT										
Train and Evaluate	Pre-train model with SSL										
	Fine-tune model with supervised learning										
	Evaluate model										
Stretch Goals	SSL with rotation prediction										

4 Evaluation

4.1 Risks

- My proposed framework may struggle to learn latent representations and be unable to perform histopathological image analysis.
- ViTs data-hungry and my chosen dataset may not meet these data requirements; causing low model accuracy and generalizability.
- My proposed model may be highly susceptible to overfitting. There are many learnable parameters, increasing the chance of the model fitting the training dataset only.
- The model may require advanced hardware during training, which I currently do not have.
- The large number of patch scales and transformer blocks may slow model inference time.

4.2 Ethical Considerations

In the case that my chosen dataset is insufficient for training my proposed model, I may have to source data from elsewhere, this includes real-world datasets. I would then need to consider how to appropriately source and process real-world datasets as they must have patient consent and be processed following GDPR requirements [34]. My chosen dataset may also include potential bias, such as class imbalance. If medical professionals employ the model, it could produce biased segmentation outputs that highlight incorrect features. Therefore, misleading medical professionals and potentially worsening patient outcomes.

5 Conclusion

In this literature review, I have analysed previous and current state-of-the-art approaches for medical image semantic segmentation, SSL, and Data Augmentation; and proposed a novel multiscale ViT-based framework based on this information. The model aims to enhance spatial and contextual information extraction in complex high-resolution images. I have provided evaluative aims and objectives for this model, which will help determine if my proposed framework successfully fulfils my hypothesis. If successful, this framework offers a potential backbone for other medical domains, such as Tumor analysis in MRI scans; professionals would only need to tweak the encoder patch sizes to best align with their domain's feature scales.

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